

MA2202

Cache

Tips

- For statements concerning subgroups of index 2, consider sgn .

Homomorphisms

- Preimages of subgroups under a homomorphism are subgroups, and preimages of normal subgroups are normal. Because homomorphisms preserve group operations, and normality is defined by conjugation, which commutes with homomorphisms.
- Injectivity at one element implies injectivity for all elements for a homomorphism.

Subgroups

- Cayley:** Every group is isomorphic to a subgroup of a symmetric group.
- Strong Cayley:** If $[G : H] = m$, there exists $\phi : G \rightarrow S_m$ and $\ker(\phi)$ is the core of H .
 - Consider the left action of G on G/H . If $gxH = xH$, $g = xhx^{-1}$ and g lies in xHx^{-1} for every $x \in G$.
- The core of a subgroup $H \leq G$
$$\text{Core}_G(H) = \bigcap_{g \in G} gHg^{-1}$$
is the largest normal subgroup of G contained in H .
- Correspondence Theorem.** If $N \trianglelefteq G$, then subgroups of G containing $N \leftrightarrow$ subgroups of G/N . The bijection is given by $G \ni H \mapsto HN$.
 - $N \subseteq H \subseteq G \iff N/N = \{e\} \subseteq H/N \subseteq G/N$

- # conjugates of $g = |G|/|C_G(g)|$.
- $|HK| = |H||K|/|H \cap K|$.
- If $G/H \leq G/K$, then $K \leq H$.
- If $N \trianglelefteq G$, G/N is simple.
- $N_G(H) = H \iff H \trianglelefteq G$.
- $N_G(H) = \{g \in G : gHg^{-1} = H\}$ is the largest subgroup of G in which H is normal (note $N_G(H)$ may not be normal in G)
- If $H \trianglelefteq G$, $K \trianglelefteq G$, and $H \cap K = \{1\}$ and $HK = G$, then $G \cong H \times K$.
- If $H \trianglelefteq G$, $K \leq G$, and $H \cap K = \{1\}$, then $HK \cong H \rtimes K$ where $\psi : K \rightarrow \text{Aut}(H)$ is given by $\psi(k)(h) = khk^{-1}$.

Commutativity

- G abelian $\iff [G, G] = \{e\}$.
- $[G, G] \trianglelefteq G$. $Z(G) \trianglelefteq G$.
- $G/[G, G]$ is the largest abelian quotient of G .
- Trivial intersection of normal subgroups forces commuting. (Show commutator in intersection).
- $G/Z(G)$ is the largest non-abelian quotient of G .
- All groups of order n are cyclic $\iff n$ squarefree and for each pair of distinct primes p and q dividing n , $q \not\equiv 1 \pmod p$

Sylow

- Tip.** Force $|N_G(P)|$ to be large for a large n_p and compare with $|G|$ for $\perp$.
- Tip.** After $G \rightarrow S_{n_p}$, either show $|G| \nmid S_{n_p}$ or the subgroup of order $|G|$ in S_{n_p} must be generated by a product of disjoint cycles of $> n_p$ length.
- $n_p = [G : N_G(P)]$.
- There is exactly one Sylow p -subgroup in $N_G(P)$, namely P .

- If G is finite simple, then $|G| \nmid [G : H]!$ for all proper subgroups H of G (Since G embeds into $S_{[G:H]}$ as the conjugation action is faithful).
 - Try: replace $[G : H]$ with $|\text{Syl}_p(G)|$
- Frattini’s Argument.** If G is a finite group, $H \trianglelefteq G$, and $P \in \text{Syl}_p(H)$, then $G = HN_G(P)$ and $[G : H] \mid |N_G(P)|$.
- Only proper non-trivial normal subgroup of S_3 is A_3 .
- Proper non-trivial normal subgroups of S_4 are A_4 , V_4 .
- For $n \geq 5$, A_n is simple.
- For $n \geq 5$, only A_n is a proper non-trivial normal subgroup of S_n .

Fermat’s little theorem

Let p be a prime number. Then
$$p \mid (n^p - n) \quad \forall n \in \mathbb{Z}$$

i.e.
$$n^p \equiv n \pmod p \implies n^{p-1} \equiv 1 \pmod p$$
Applying this idea,
$$n^{a(p-1)+b} \equiv n^b \pmod p$$

Functions

- Let A, B be sets, and let $f : A \rightarrow B$ be a function.
- Let $B' \subseteq B$. Define
$$f^{-1}(B') = \{a \in A : f(a) \in B'\}$$
 - If $S \subseteq A$, then $f|_S$ denotes the same function except that the domain A is replaced by S . This function $f|_S$ is called the restriction of f to S .
 - The inclusion fn on Y for some $Y \subset X$ is the function $h : Y \rightarrow X$ defined by $h(y) = y \forall y \in Y$.
 - A fn is a bijection $\iff$ its inverse fn exists.

Group isomorphisms

- Definition**
- If $G \cong H$:
- If G is abelian, H is abelian.
 - $\phi : G \rightarrow G$ given by $\phi(g) = g^{-1}$ is a group isomorphism $\iff G$ is an abelian group.

Two isomorphisms

Suppose $\phi : (G, *) \rightarrow (H, \star)$ and $\psi : (H, \star) \rightarrow (K, \cdot)$ are two isomorphisms of groups. Then

- the inverse function $\phi^{-1} : (H, \star) \rightarrow (G, *)$ and
- the composite function $\psi \circ \phi : (G, *) \rightarrow (K, \cdot)$

are group isomorphisms.

Subgroups

- Roots of unity** $(\mu_m, \times)$ is a subgroup of $(\mu_n, \times)$ if $m \mid n$.
- Let $(G, *)$ be a group and let $H \subseteq G$ be a nonempty subset. Then $(H, *)$ is a subgroup iff for all $a, b \in H$, we have $a * b^{-1} \in H$.
 - Intersection of a finite collection of subgroups is a non-empty subgroup.
 - Let $(H, *)$ and $(K, *)$ be subgroups of $(G, *)$. If $(H \cup K, *)$ is a subgroup, then either $H \subseteq K$ or $K \subseteq H$.
 - Subgroups of cyclic groups are cyclic.

Cycles

- $(i_1 i_2 \cdots i_r) = (i_1 i_r)(i_1 i_{r-1}) \cdots (i_1 i_2)$.

- Alternating group** Let
$$A_n = \{f \in S_n : \text{sgn}(f) = 1\} = \{f \in S_n : f \text{ even}\}$$
be the set of all even permutations in S_n . Then $(A_n, \circ)$ is a subgroup of $(S_n, \circ)$.
- The subset of odd permutations is not a subgroup.
 - All k -cycles in S_n are conjugate but not necessarily in A_n

Cayley’s theorem

Let $(G, *)$ be a finite group of order n . Then $(G, *)$ is isomorphic to a subgroup of $(S_n, \circ)$.

Proof

- We know that $(S_Y, \circ)$ is isomorphic to $(S_n, \circ)$.
- Let $Y = G$. For every $g \in G$, define function $f_g : Y \rightarrow Y$ by
$$f_g(y) = g * y \text{ for all } Y = G$$
Then construct $\phi : G \rightarrow S_Y$ by $\phi(g) = f_g$. ϕ is an injective group homomorphism, so G is isomorphic to the image G' which is a subset of S_Y , i.e. G is isomorphic to a subgroup of $(S_Y, \circ)$.

Cosets and Lagrange’s theorem

- If G is abelian, then a left coset is also a right coset.
- If $z \in xH$, then $zH = xH$.
- If $xH \cap yH \neq \emptyset$, then $xH = yH$.
- The collection of left cosets $\{xH : x \in G\}$ forms a partition of G .
- Every coset xH is of the same cardinality as H .
- If H is a subgroup of G and K is a subgroup of H , then $[G : K] = [G : H][H : K]$

Generators

- Subgroup** Let G be a group, and let $X \subseteq G$. Let $S = \{H : H \text{ subgroup of } G, H \supseteq X\}$. We define
$$\langle X \rangle = \bigcap_{H \in S} H$$
- $\langle X \rangle$ is the smallest subgroup of G containing X .

Group homomorphisms

- Let $\phi : (G, *) \rightarrow (H, \star)$ be an injective group homomorphism. Then $(G, *)$ is isomorphic to its image which is a subgroup of $(H, \star)$.
- Let H' be a subgroup of H . Then $\phi^{-1}(H')$ is a subgroup of G .
- ϕ is injective $\iff \ker \phi$ is trivial.

Group homomorphisms and subgs

- Let $(G, *)$ and $(H, \star)$ be two groups and let $\phi : (G, *) \rightarrow (H, \star)$ be a group homomorphism. Let $K = \ker \phi$. Define
- Sub** $(G, K) = \{G' : G' \text{ subgroup of } G, G' \supset K\}$ which contains all the subgroups of G which contain K and
 - Sub** $(H) = \{H' : H' \text{ subgroup of } H\}$
- Define a function $\Phi : \text{Sub}(G, K) \rightarrow \text{Sub}(H)$ by $\Phi(G') = \phi(G')$ where $G' \in \text{Sub}(G, K)$. Then, $\phi(G')$ is a subgroup of H , so $\Phi(G') \in \text{Sub}(H)$.

Theorem 46 Suppose ϕ is a surjective homomorphism. Then Φ is a bijection.

Normal subgroups

- If G is abelian, every subgroup is normal.
 - Converse is not true. Q_8 is not abelian but every subgroup is normal.
- Center of a group is a normal subgroup.
- K normal $\iff$ for all $g \in G$, $(gK)(g'K) = (gg')K$.

Simple groups

- Let p be a prime number. $\mathbb{Z}/p\mathbb{Z}$ is a simple group.
- For $n \neq 4$, the alternating group A_n is a simple group.
- Lemma** Let H be a normal subgroup of A_n where $n \geq 5$. If H contains a 3-cycle, $H = A_n$.
- H contains a 3-cycle $\implies H$ contains all the 3-cycles of A_n . Every even permutation is the product of 3-cycles. Hence $H = A_n$.
- Lemma** $A_{n-1} \leq A_n$ by $A_{n-1} = \{\sigma \in A_n : \sigma(n) = n\}$.

Lemma Let $S \leq G$ and $N \trianglelefteq G$. The following hold:

- $SN \leq G$. (If $S \trianglelefteq G$, then $SN \trianglelefteq G$).
- $N \trianglelefteq SN$.
- $S \cap N \trianglelefteq S$.
- Quotient groups $(SN)/N \cong S/(S \cap N)$.

Technically, it is not necessary for N to be a normal subgroup, as long as S is a subgroup of the normalizer of N in G . In this case, N is not a normal subgroup of G , but N is still a normal subgroup of the product SN .

Quotient groups

Theorem

- The function $\pi : G \rightarrow G/K$ defined by $\pi(g) = gK$ for all $g \in G$ is a surjective group homomorphism.
- The kernel of π is K .

The First Isomorphism Theorem

In this section, $(G, *)$ and $(H, \star)$ are (possibly infinite) groups. Let $\phi : (G, *) \rightarrow (H, \star)$ be a surjective group homomorphism. Let K be the kernel of ϕ .

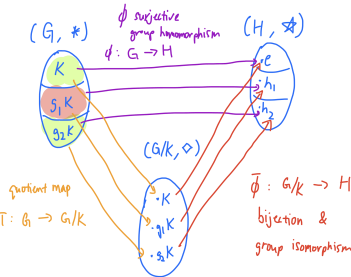
- Suppose $\phi(g) = h$ where $h \in H$ and $g \in G$. Then $\{x \in G : \phi(x) = h\} = gK$
i.e. the whole of gK is sent to h under ϕ .

First Isomorphism Theorem

Let $\phi : (G, *) \rightarrow (H, \star)$ be a surjective group homomorphism. Let K be the kernel of ϕ . Then the function $\bar{\phi} : (G/K, \diamond) \rightarrow (H, \star)$ given by

$$\bar{\phi}(gK) = \phi(g)$$

is a well-defined group isomorphism.



- If ϕ is not surjective, then replace H with the image $H' = \phi(G)$ in the definition of $\bar{\phi}$.

Corollary Let $\phi : G \rightarrow H$ and $\psi : G \rightarrow H'$ be two group homomorphisms.

- Suppose ϕ and ψ have the same kernel K . Then, the images $\phi(G)$ and $\psi(G)$ are isomorphism groups.

- If G is a finite group, then

$$|\phi(G)| = |\psi(G)| = |G/K| = |G| / |K|$$

The Second Isomorphism Theorem

In this section, G is a group, M is a subgroup of G , and K is a normal subgroup of G .

Prop. 59 $MK = KM$ and it is a subgroup of G .

Proposition 60

- The function $\phi : M \rightarrow MK/K$ defined by $\phi(m) = mK$ is a surjective group homomorphism.
- The kernel of ϕ is $M \cap K$. In particular, it is a normal subgroup of M .

Second Isomorphism Theorem

$$M/(M \cap K) \simeq (MK)/K$$

The Third Isomorphism Theorem

Let G be a group. Let M and K be normal subgroups of G such that $M \supseteq K$. Then M/K is a normal subgroup of G/K and

$$(G/K)/(M/K) \simeq G/M$$

If $M \not\supseteq K$, then replace K by $M \cap K$, which is a normal subgroup of G contained in M .

Corollary Let M and K be normal subgroups of G such that $M \supseteq K$. Then there is a surjective group homomorphism

$$\phi : G/K \rightarrow G/M$$

given by $\phi(gK) = gM$.

Euler's totient function

Let n be a positive integer. If $n = 1$, set $\Phi(1) = \{1\}$. Else set

$$\Phi(n) = \{x \in \mathbb{Z} : 0 \leq x \leq n, \gcd(x, n) = 1\}$$

- Let $*$ denote multiplication modulo n . Then $(\Phi(n), *)$ is a group.
- Let $\phi(n)$ denote the number of elements in $\Phi(n)$.
- For prime number p , $\Phi(p) = \{1, 2, \dots, p-1\}$, so $\phi(p) = p-1$.
- For prime number p , let $n = p^r$.

$$\phi(p^r) = n - \frac{n}{p} = p^r \left(1 - \frac{1}{p}\right) = p^{r-1}(p-1)$$

Euler's theorem

Let x be an integer such that $\gcd(x, n) = 1$. Then

$$x^{\phi(n)} \equiv 1 \pmod{n}$$

Calculating $\phi(n)$

Suppose $n = p_1^{r_1} p_2^{r_2} \dots p_n^{r_n}$. Then

$$\begin{aligned} \phi(n) &= n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right) \\ &= \phi(p_1^{r_1}) \phi(p_2^{r_2}) \dots \phi(p_k^{r_k}) \end{aligned}$$

Automorphism groups

Let $(G, *)$ be a group. An isomorphism $\phi : G \rightarrow G$ is called an automorphism of G . We denote the set of automorphisms of G by

$$\text{Aut}(G) = \{\phi : G \rightarrow G : \phi \text{ is an isomorphism}\}$$

Isomorphism facts

- Identity map id_G is an isomorphism.
- Composition of isomorphisms is an isomorphism, i.e. $\circ$ is a binary operation on $\text{Aut}(G)$.
- Inverse of an isomorphism is an isomorphism.

Proposition $(\text{Aut}(G), \circ)$ forms a group.

- A subgroup A of $(\text{Aut}(G), \circ)$ is called an automorphism subgroup.

Inner automorphism

$\phi_g : G \rightarrow G$ given by $\phi_g(x) = gxg^{-1}$ is called an inner automorphism of G .

- Let $\text{Inn}(G) = \{\phi_g : g \in G\}$ be the set of inner automorphisms.
- The subset $\text{Inn}(G)$ is a normal subgroup of $(\text{Aut}(G), \circ)$.

Proposition The map $T : G \rightarrow \text{Inn}(G)$ given by $T(g) = \phi_g$ is a surjective group homomorphism whose kernel is the center of the group

$$Z(G) = \{z \in G : gz = zg \text{ for all } g \in G\}$$

By the first isomorphism theorem,

$$G/Z(G) \simeq \text{Inn}(G)$$

The Sylow Theorems

First Sylow Theorem

Let G be a group of order n . Let p be a prime divisor of n . Then G contains a Sylow p -subgroup.

Corollary Let G be a finite group of order n . Let p be a prime divisor of n . If $p^d | n$, then G contains a subgroup of order p^d .

Theorem 94

Let G be a group of order n . Let $\{P_1, P_2, \dots, P_r\}$ be all the distinct conjugates of a Sylow p -subgroup $P = P_1$.

- Let Q be a p -subgroup of G . Then $Q \subseteq P_i$ for some i .
- (Second) If Q is a Sylow p -subgroup of G , then $Q = P_i$ for some i .
- (Third) Let r denote the number of Sylow p -subgroups of G . Then
$$r \equiv 1 \pmod{p} \text{ and } r | [G : P]$$

Corollary 95 Let P be a Sylow p -subgroup of a finite group G . Then P is a normal subgroup $\iff P$ is the unique Sylow p -subgroup of G .